

# Constructed Exposure Measures and Statement-Specific Robustness: Evidence from Early-Career Employment

Lily Luo

[AUTHOR TO COMPLETE: affiliation]

[AUTHOR TO COMPLETE: email]

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**Abstract.** Occupational AI exposure is constructed before it enters a regression. I hold a Current Population Survey employment-stock design fixed and vary technology scope, source labels, aggregation, taxonomy mapping, support, and treatment representation. Taxonomy repair raises computer-and-mathematical employment coverage from 3.33 to 97.7 percent, mainly by readmitting occupations rather than changing scores on fixed support. Within two selected exposure families, six implementations yield negative high-versus-low point estimates on common support, but a broader architecture audit changes the conclusion: Webb AI estimates a smaller, imprecise contrast and an OECD capability measure estimates nearly zero. The original  $-0.1311$  log-point beta contrast is also reference-dependent—Q5 differs from Q1 and Q3, but not detectably from Q2 or Q4—and a broad-family permutation diagnostic does not establish AI specificity. Ranking disagreement is common but scale-sensitive: six-way switch conflict is 53.28 percent at a zero threshold, 40.10 percent when weighted by movement mass, and substantial opposition falls to 14.00 percent at a 0.5-standard-deviation threshold. The evidence supports a measurement-centered conclusion: robustness belongs to a specified architecture, support, reference group, and economic statement, not to the label “AI exposure” in general.

**JEL codes:** J23, J24, O33, C81.

**Keywords:** artificial intelligence, occupational exposure, measurement, early-career employment, constructed treatments.

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# 1 Introduction

An occupational “AI exposure” variable is already an empirical model before it reaches a regression. Its authors have chosen a technology boundary, an occupational primitive, a source of labels, and an aggregation rule. A downstream researcher then chooses a taxonomy bridge, support rule, normalization, conditioning set, and treatment representation. These choices can change both the score and the economic comparison it defines.

This paper asks which of those choices alter which conclusions. Robustness is statement-specific. A high-versus-low coefficient consumes tail membership and a reference group. A continuous regression consumes residual variation after controls. An occupational-mobility claim consumes pairwise ranks and a threshold for an economically meaningful move. Agreement for one object does not imply agreement for the others.

I use early-career employment as a demanding application. The fixed design compares the post-January-2023 evolution of CPS employment stocks for ages 22–25 and 26–65 across occupational exposure groups. It was originally motivated by the age gradient in private payroll data documented by [Brynjolfsson, Chandar, and Chen \(2026\)](#). The CPS analysis does not observe AI adoption, and its occupation-level timing is not an instrument. Its role here is to stress-test a constructed treatment while holding the outcome and estimator fixed.

The architecture audit begins with two selected families and six implementations: three ability-based AIOE variants ([Felten, Raj, and Seamans 2018, 2021](#)), and direct, software-complemented, and broad task-based LLM measures ([Eloundou et al. 2024](#)). These are not six independent observations of one known latent exposure. I then admit two outside architectures under an outcome-independent rule: Webb’s AI patent–task measure and the OECD capability-gap measure ([Webb 2020](#); [Organisation for Economic Co-operation and Development 2026](#)).

Five results determine the revised framing. First, taxonomy repair raises computer-and-mathematical employment coverage from 3.33 to 97.7 percent. On fixed support it barely changes a continuous AIOE coefficient, from  $-0.01885$  to  $-0.01920$ ; most of the change to  $-0.03156$  comes from readmitting occupations. Second, the six selected implementations all give negative common-support tail point estimates, but this does not extend to the wider architecture set: Webb AI gives  $-0.0649$  with an interval spanning zero, and the OECD measure gives  $-0.0110$ .

Third, the original beta tail estimate of  $-0.1311$  log points is reference-dependent. Q5 differs from Q1 and Q3, but its differences from Q2 and Q4 are much smaller and not detected. Separate Q1 and Q5 stock paths and deletion checks show that favorable low-tail evolution contributes to the contrast. Fourth, the beta estimate is more negative than

education, cognitive-content, and telework benchmarks on identical support, but a within-broad-occupation permutation diagnostic does not place it unusually far in the lower tail. AI specificity is therefore not established.

Fifth, occupational ranking disagreement is real but economically scale-dependent. At a zero threshold, 53.28 percent of linked occupational switches receive opposite directions from at least two of the six measures. Weighting conflict by movement mass lowers the share to 40.10 percent; requiring substantial opposition of at least 0.5 standard deviations lowers it to 14.00 percent. Yet each architecture classifies roughly half of aggregate switch weight as moving upward. Architecture choice is important for marginal rankings without overturning the aggregate direction of reallocation.

The contribution is a linked measurement audit. It follows construction through taxonomy coverage, support, tail membership, estimator information, employment contrasts, and mobility statements. This complements sensitivity work that asks which moments drive estimates (Andrews, Gentzkow, and Shapiro 2017) and work on uncertainty in trend-based designs (Rambachan and Roth 2023), without borrowing identification that the CPS design does not possess. It also preserves the paper’s design record: a predeclared all-six simultaneous-band criterion failed, and realized standard errors were more than three times prospective calculations.

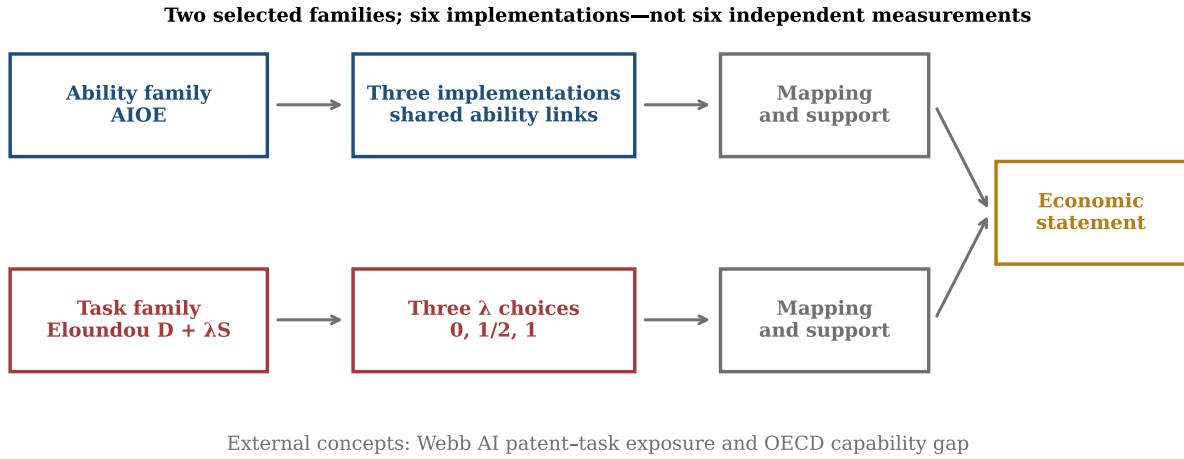
Figure 1 summarizes the architecture chain. The remainder of the paper defines the constructs, documents the data and estimator, reports employment stress tests and continuous-family models, and then examines mobility before synthesizing what the application can and cannot establish.

## 2 Constructs, Architecture Space, and Related Literature

### 2.1 Two selected families

AIOE links progress in AI applications to O\*NET abilities and combines those links with occupational ability requirements (Felten, Raj, and Seamans 2018, 2021). The three implementations used here share that conceptual system and differ mainly in how scores are reconstructed and transported across taxonomies. Their employment-weighted level correlations range from 0.981 to 0.996. They are dependent implementations, not independent replications.

The Eloundou family begins from occupational tasks and a time-saving rubric (Eloundou et al. 2024). Let  $D_o$  denote the share directly accelerated by an LLM and  $S_o$  the share



**FIGURE 1.** From exposure construction to an economic statement

*Notes:* The six initial implementations belong to two selected families. Mapping and support are downstream choices. Webb AI and OECD capability exposure enter as outside architectures under a rule fixed before their outcome models were estimated. Exposure measures potential susceptibility, not realized adoption.

accelerated only with complementary software. Before separate standardization,

$$X_{\alpha o} = D_o, \quad X_{\beta o} = D_o + \frac{1}{2}S_o, \quad X_{\gamma o} = D_o + S_o. \quad (1)$$

On the 444-occupation strict common support, common mapping and aggregation preserve  $X_\beta = (X_\alpha + X_\gamma)/2$  to a maximum absolute error below  $7.0 \times 10^{-13}$ . Separately standardizing the three variables breaks the numerical midpoint identity—the maximum gap is 0.5112—but creates no new primitive dimension. I therefore never enter alpha, beta, and broad together as independent primitive regressors.

Within the task family, I evaluate  $X(\lambda) = D + \lambda S$  for  $\lambda \in \{0, .25, .5, .75, 1\}$  while holding mapping, support, and other design choices fixed. Fixed-beta-scale and restandardized continuous slopes are reported separately. Categorical tail estimates range from  $-0.1013$  to  $-0.1482$  across the grid; they need not vary mechanically with the continuous slopes because quintile membership changes.

## 2.2 Outside architectures and prior computerization

An architecture was admitted only if its construct was documented, its source was public and hashed, mapping did not rely on title matching, full components were available, quintile cuts were distinct, and preperiod employment coverage exceeded 80 percent. Webb AI patent–task overlap and the reversed OECD capability gap passed this rule. Webb software exposure remains a conditioning variable: its source correlation with Webb AI is 0.7021, substantial but far from identity. Frey–Osborne is retained as an automation-risk comparator rather than relabeled as an AI architecture. No additional LLM-labeled score was admitted after outcomes were seen.

This distinction also addresses computerization. Webb software, O\*NET computer-use measures, routine-task intensity, and automation risk are not interchangeable controls. They remove different occupational comparisons. The design can ask how an AI-exposure association changes after conditioning on each; it cannot isolate a causal AI shock from a decades-long computerization process merely by adding a correlated index.

## 2.3 Relation to labor-market evidence

The most comparable descriptive benchmark is [Brynjolfsson, Chandar, and Chen \(2026\)](#), who use ADP payroll data through June 2026 and report a 19-percent gap for ages 22–25 in exposed occupations, operating mainly through hiring. Their revised paper examines alternative exposure measures and competing shocks and labels the evidence descriptive. The

present application uses public CPS stocks, a different occupation crosswalk, and a fixed young-versus-older estimand. It does not reproduce the ADP firm or hiring margins.

[Humlum and Vestergaard \(2025\)](#) instead link representative adoption surveys to Danish administrative records and estimate effects on earnings and recorded hours. Their country, occupations, outcomes, observed adoption, design, and period differ from both ADP and the present application. Changing exposure construction within a U.S. CPS stock design cannot decompose the ADP–Denmark disagreement. Workplace deployment studies such as [Brynjolfsson, Li, and Raymond \(2025\)](#) make the exposure/adoption distinction especially clear: observed tool access can affect productivity even when an occupation-level potential-exposure score is held fixed.

The broader task literature emphasizes that technologies substitute for some tasks, complement others, and induce occupational adjustment ([Autor, Levy, and Murnane 2003](#); [Autor and Dorn 2013](#); [Acemoglu and Restrepo 2019](#)). Work on generated covariates shows why ordinary regression uncertainty generally does not absorb label-generation uncertainty ([Battaglia et al. 2024](#); [Christensen and Hansen 2026](#); [Duan and Pelger 2026](#); [Ludwig, Mullainathan, and Ramachan 2026](#)). I do not observe a validated true exposure, so the paper audits architectures rather than estimating a corrected latent variable.

A targeted search of the AER, QJE, JPE, Econometrica, Review of Economic Studies, Review of Economics and Statistics, AEJ: Applied, Journal of Finance, Review of Financial Studies, and Journal of Financial Economics found adjacent work on AI deployment, automation, firm investment, remote work, and knowledge production. It did not locate a peer-reviewed article in those ten journals that fixes this CPS young-versus-older stock design while tracing exposure construction through taxonomy, reference tails, estimator information, and mobility. This is a scoped search result, not a universal priority claim.

## 3 Data, Calendar, Taxonomy, and Samples

### 3.1 CPS employment stocks and calendar repair

The public-data application uses IPUMS basic monthly CPS records from January 2017 through July 2026. Employed respondents ages 18–65 with positive final weights are aggregated into occupation-by-age-group-by-month stocks. The headline sample defines young workers as ages 22–25 for comparability to [Brynjolfsson, Chandar, and Chen \(2026\)](#); ages 18–21 enter a sensitivity. The outcome is an employment stock, not an individual employment probability. It can change through entry, exit, occupational switching, or the older comparison stock.

The original wide extract accidentally requested the March ASEC sample ('03s') rather than the basic sample ('03b') in 2017–2021. The ASEC observations have zero basic monthly final weights and are unusable for this design. A separately hashed repair restores the five basic March samples while preserving the frozen extract. October 2025 is categorically different: no CPS was collected during the federal government shutdown. The corrected raw files contain 9,843,021 records and 114 observed survey months. The frozen static model has 108 months because it also excludes transition month December 2022; the corrected-calendar sensitivity has 113.

Restoring the March samples changes the primary coefficient from  $-0.1311$  to  $-0.1346$  (95-percent interval  $[-0.2230, -0.0461]$ ). The calendar omission does not explain the result. An attempted occupation-by-age-by-month-of-year sensitivity, adding 5,148 nuisance parameters, did not converge and is reported as a failed diagnostic.

### 3.2 Taxonomy repair

The CPS uses Census 2010 occupation codes through 2019 and Census 2018 codes thereafter, while source exposures use SOC vintages. Pre-2020 records are route-expanded using official conversion proportions, then mapped to the common target taxonomy. In one-to-many cases, the same source conversion weights apply to young and older workers. This mechanically preserves a source occupation's young/older ratio across its target components before those targets receive other sources. Fifty-six source codes are one-to-many, maximum multiplicity is seven, and they represent 20.03 percent of early-period weighted employment.

This bridge matters most through support. A literal AIOE SOC-2010 to post-2018 exact-code merge covers only 3.33 percent of computer-and-mathematical employment; the documented repair covers 97.7 percent. In a separate continuous decomposition, repairing scores on the original 410-occupation support changes the coefficient from  $-0.01885$  to  $-0.01920$ . Expanding to 495 occupations changes it to  $-0.03156$ ; excluding computer/math yields  $-0.02940$ . The main effect is re-admission, not score correction among already matched occupations. Exact-code matching across incompatible vintages is therefore documented as a failure mode, not treated as a defensible preferred estimate.

Two alternative harmonizations bound the bridge. Restricting to the Census-2018 period produces  $-0.1211$  on 457 occupations and 77 months; aggregating to a stable Census-2010 taxonomy gives  $-0.1522$  on 408 occupations and 113 months. The former changes the temporal estimand and the latter sacrifices detail.

### 3.3 Sample accounting and weighting

Table 1 reconciles four universes that appeared separately in earlier versions. The 490 candidate occupations are narrowed by treatment-side characteristic availability (463), primary beta plus Webb support (468), beta plus O\*NET computer-use support (465), or literal six-measure plus Webb support (444). The common-support employment percentages are relative to the 490-occupation candidate stock, not claims of unrestricted national coverage.

**TABLE 1.** Occupation universes and employment coverage

| Universe                  | Occ. | Nonempty cells | Young resp.-eq. | Older resp.-eq. | Post young cov. | Post older cov. |
|---------------------------|------|----------------|-----------------|-----------------|-----------------|-----------------|
| Characteristics           | 463  | 48,742         | 314,617         | 3,687,156       | 94.80%          | 94.04%          |
| Primary beta + Webb       | 468  | 49,579         | 324,522         | 3,812,564       | 98.53%          | 97.96%          |
| Beta + O*NET computer use | 465  | 48,909         | 313,098         | 3,706,485       | 94.02%          | 94.52%          |
| Six measures + Webb       | 444  | 47,073         | 311,223         | 3,646,830       | 93.89%          | 93.08%          |

Notes: Each universe covers 108 static-model months. Coverage is relative to the 490-occupation candidate stock separately by age group and period. Respondent-equivalent counts use CPS weights divided by their scale after pre-2020 route expansion and are not unique people. The primary weighted young and older stocks are 1.161 and 11.842 billion occupation-month units. Full pre/post coverage and exclusion counts appear in the online appendix.

On primary support, Q1 contains 133 occupations and 20.02 percent of employment; Q5 contains 91 and 19.53 percent. Ties are retained. Primary quintile cutoffs and score normalizations use preperiod employment weights. Every support sensitivity states whether cutoffs and scales are frozen or recomputed. Respondent-equivalent counts after route expansion are not unique respondents.

## 4 Estimands, Design, and Inference

### 4.1 Headline objective

Let  $N_{oat}$  denote the CPS-weighted employment stock in occupation  $o$ , age group  $a$ , and month  $t$ . Let  $Y_a$  identify ages 22–25,  $P_t$  identify January 2023 onward,  $Q_{oq}$  denote exposure quintile  $q$ , and  $W_o$  denote standardized Webb software exposure. The conditional mean is

$$E[N_{oat}|X] = \exp \left[ \gamma_{oa} + \delta_{ot} + \lambda_{at} + \sum_{q=2}^5 \beta_q Q_{oq} Y_a P_t + \theta W_o Y_a P_t \right]. \quad (2)$$

The fixed effects are occupation-by-age, occupation-by-month, and age-by-month. The target  $\beta_5$  is the post-period change in the young-to-older employment-stock ratio for Q5 relative



to Q1, conditional on Q2–Q4 and Webb. It is neither a continuous exposure slope nor an individual probability effect.

With two age groups, conditioning independent Poisson pseudo-likelihood contributions for  $N_{oyt}$  and  $N_{oot}$  on  $T_{ot} = N_{oyt} + N_{oot}$  yields a grouped-binomial contribution

$$\ell_{ot} = N_{oyt} \log p_{ot} + N_{oot} \log(1 - p_{ot}), \quad \text{logit}(p_{ot}) = \log \frac{\mu_{oyt}}{\mu_{oot}}. \quad (3)$$

Occupation-month terms cancel from the ratio; the remaining fixed effects and interactions reproduce the implemented objective. CPS person weights construct the two cell totals. Those totals are estimating-equation inputs, not literal counts of independent survey-weighted binomial trials, and they receive no second likelihood weight. The six-bin age profile is multinomial and is not justified by this binary derivation.

The primary sample has 50,544 potential occupation-month cells, of which 13,305 have zero young stock, 1,036 zero older stock, and 965 both zero. Estimation retains 49,579 nonempty cells. Minimum full-panel respondent-equivalent thresholds of 100, 250, and 500 produce coefficients between  $-0.1311$  and  $-0.1326$ . A separately constructed unweighted respondent-equivalent outcome gives  $-0.1308$ ; it changes the cell estimand and does not double-weight CPS stocks.

## 4.2 Inference and information

The frozen inference clusters by occupation and uses 999 common occupation-level Rademacher wild-score multipliers. Shared multipliers preserve covariance for paired architecture and reference-group contrasts. A two-way occupation/calendar-month model-based sensitivity changes the primary SE only from 0.0444 to 0.0449; neither is a complete CPS survey-design correction.

Four information objects must remain separate. The primary model has 468 nominal occupation clusters. Webb-residualized beta variation has inverse-Herfindahl effective support 53.26 and a top-five share of 22.15 percent. Fitted conditional information has effective count 43.30 and top-five share 24.57 percent. Actual leave-one-occupation-out movement is at most 0.0205 and never flips the coefficient sign. Effective counts are concentration summaries, not numbers of independent clusters.

The all-six sign statement illustrates compound inference. For

$$H_0 : \text{at least one } \beta_m \geq 0, \quad H_1 : \beta_m < 0 \text{ for every } m,$$

the marginal one-sided tail areas are .045, .005, .006, .008, .001, and .001, so the intersection-union  $p$ -value is their maximum, .045. The predeclared simultaneous one-sided band is a different deliverable: its critical value is 2.26036 and the limiting administrative-AIOE upper bound is +0.01858. That criterion fails and remains visible.

Realized uncertainty also exceeded planning calculations. Primary and paired realized/prospective SE ratios are 3.649 and 3.167 on matched estimands, support, and units. The planning simulations omitted contemporaneous cross-occupation covariance, survey noise, new post shocks, structural change, measurement error, and misspecification. Existing artifacts cannot identify a unique allocation of the gap. A literal residual-wild full refit was attempted for the primary and no-alpha F/G models, but all 199 pseudo-outcomes violated grouped-binomial bounds; no clipping or different bootstrap was substituted.

Confirmatory choices were frozen before protected outcomes were opened. Common-support, age, calendar repair, placebo, external-architecture, family-refit, and mobility audits are post-outcome exploratory. This chronology limits claims but does not conceal important failures.

## 5 Employment Application: Stress Tests

### 5.1 Baseline, architecture scope, and specificity

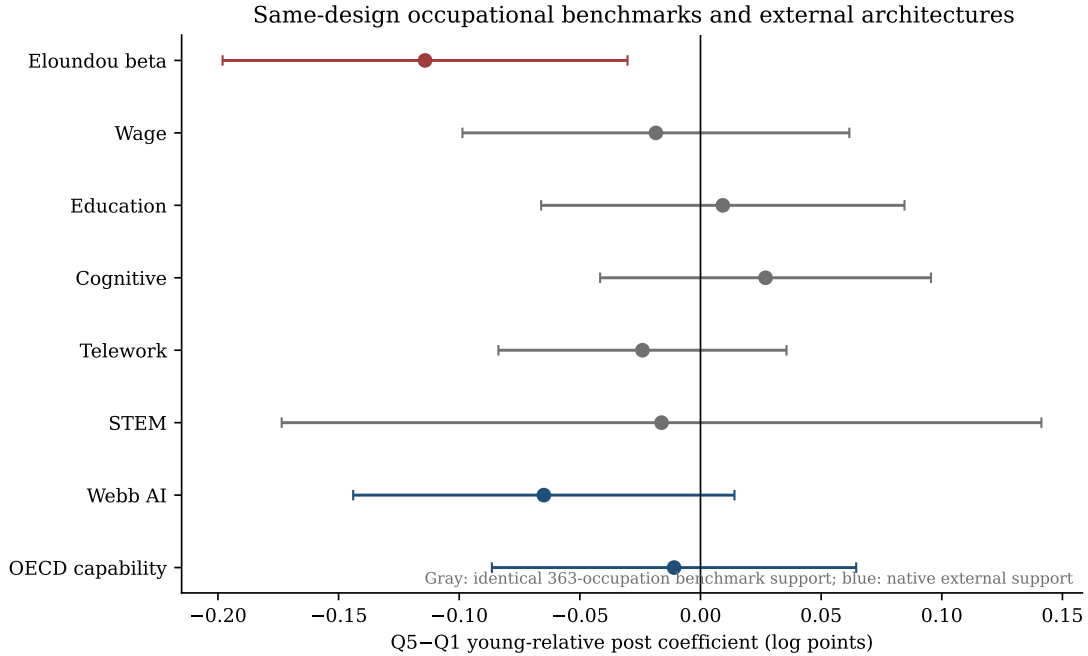
The frozen primary beta-by-Webb Q5–Q1 estimate is  $-0.1311$  log points (occupation-cluster SE 0.0444; wild-score interval  $[-0.2170, -0.0451]$ ). The monotone transformation  $100[\exp(\hat{\beta}) - 1]$  is  $-12.28$  percent; transformed endpoints are  $-19.50$  and  $-4.41$  percent. These are relative-stock-ratio changes, not employment-probability changes, job counts, or percentage points.

On the identical 444-occupation intersection, the three AIOE point estimates are  $-0.0739$ ,  $-0.1029$ , and  $-0.1021$ ; the task-family estimates are  $-0.1013$ ,  $-0.1290$ , and  $-0.1465$ . All are negative, but the predeclared simultaneous-band criterion fails as described in Section 4. This sign pattern is within two selected, internally dependent families.

The external-architecture audit is less supportive. Webb AI gives  $-0.0649$  with interval  $[-0.1439, 0.0141]$  on 448 occupations. The reversed OECD capability gap gives  $-0.0110$  with interval  $[-0.0864, 0.0645]$ . On common supports, beta is more negative by 0.0646 and 0.1115, respectively; both paired intervals include zero, the OECD comparison narrowly so. The six-measure sign statement therefore does not generalize to every admissible exposure concept.

Figure 2 adds generic occupational-characteristic benchmarks. On identical 363-occupation support, beta is  $-0.1142$ . Wage, education, cognitive content, natural telework groups, and

STEM status give  $-0.0185$ ,  $+0.0093$ ,  $+0.0270$ ,  $-0.0240$ , and  $-0.0161$ . Common-draw beta-minus-benchmark intervals exclude zero for education, cognition, and telework, but not wage or STEM. Telework and STEM cannot form five tie-preserving groups, so the declared natural groups replace quintiles rather than arbitrary tie-breaking.



**FIGURE 2.** Same-design benchmarks and external architectures

*Notes:* Red and gray rows use identical 363-occupation support. Telework and STEM use natural groups because tie-preserving quintiles collapse. Blue rows use each external architecture’s admitted native support and should not be read as a single common-sample panel. Lines are 95-percent one-step occupation-cluster wild-score intervals. All rows are post-outcome exploratory.

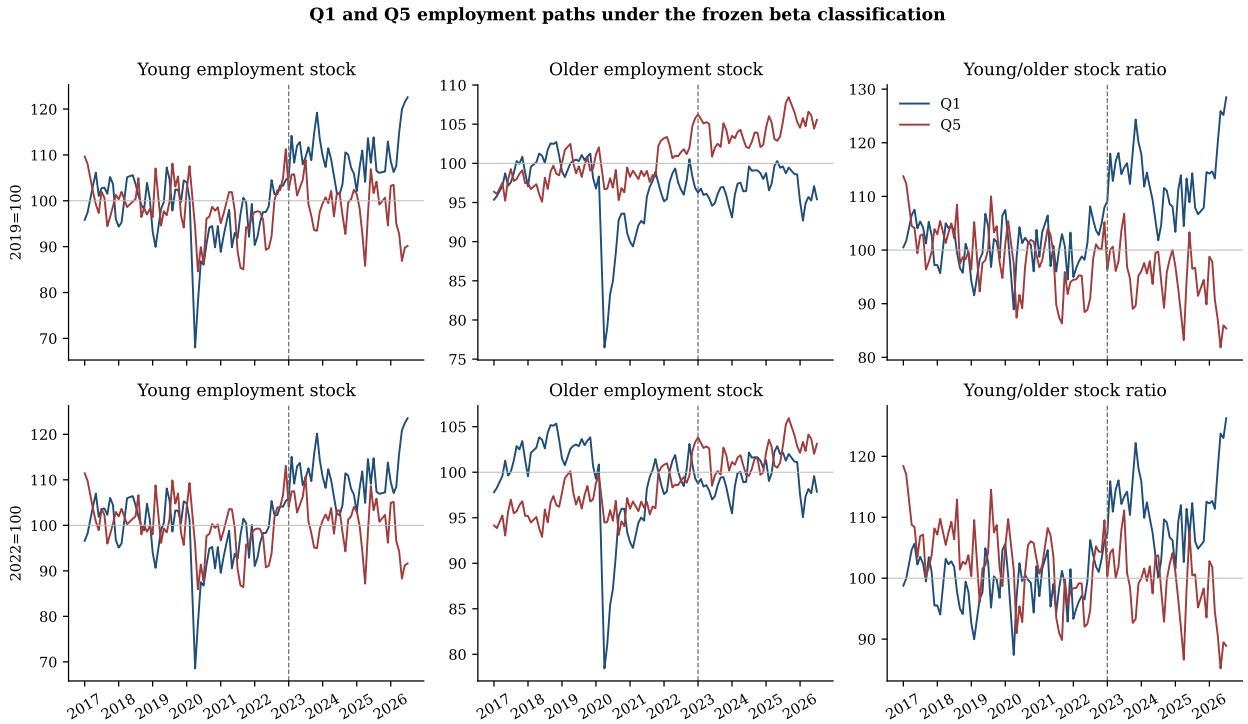
A separate diagnostic permutes beta scores within two-digit SOC families 999 times, preserving each family’s score multiset and the outcome design while reassigning detailed occupations. All fits converge. The observed coefficient is  $-0.1311$  versus a permutation mean of  $-0.1097$  and SD  $0.0268$ ; its lower-tail probability is  $.219$  and centered two-sided probability  $.434$ . Because occupational score exchangeability within families is not justified, this is not randomization inference. Together, the placebos and permutation support a bounded statement: the beta contrast is more negative than several selected characteristics, but AI specificity is not established.

## 5.2 Which tail supplies the contrast?

The full beta profile relative to Q1 is  $-0.0855$ ,  $-0.0478$ ,  $-0.0970$ , and  $-0.1311$  for Q2 through Q5. The Q5–Q2 estimate is  $-0.0456$  with interval  $[-0.1240, 0.0328]$ ; Q5–Q4 is  $-0.0340$

with interval  $[-0.1141, 0.0460]$ ; Q4–Q2 is  $-0.0116$  with interval  $[-0.0725, 0.0494]$ . Q5 differs detectably from Q3 ( $-0.0833$ , interval  $[-0.1582, -0.0083]$ ), but not from Q2 or Q4. The headline therefore depends partly on Q1 rather than revealing a monotone dose response.

Figure 3 plots young stocks, older stocks, and their ratios separately for Q1 and Q5, normalized to observed 2019 and 2022 means. It confirms that the fitted contrast cannot be narrated as only a collapse in high-exposure young employment. The low-exposure reference follows its own post-pandemic path, and older stocks contribute to both ratios. These paths are descriptive and do not additively decompose a nonlinear fitted coefficient.



**FIGURE 3.** Q1 and Q5 employment stocks and ratios

*Notes:* Monthly CPS-weighted stocks under frozen beta membership. The top row indexes each series to its mean over observed 2019 months; the bottom row uses observed 2022 months. The dashed line marks January 2023. October 2025 is absent because no CPS was collected.

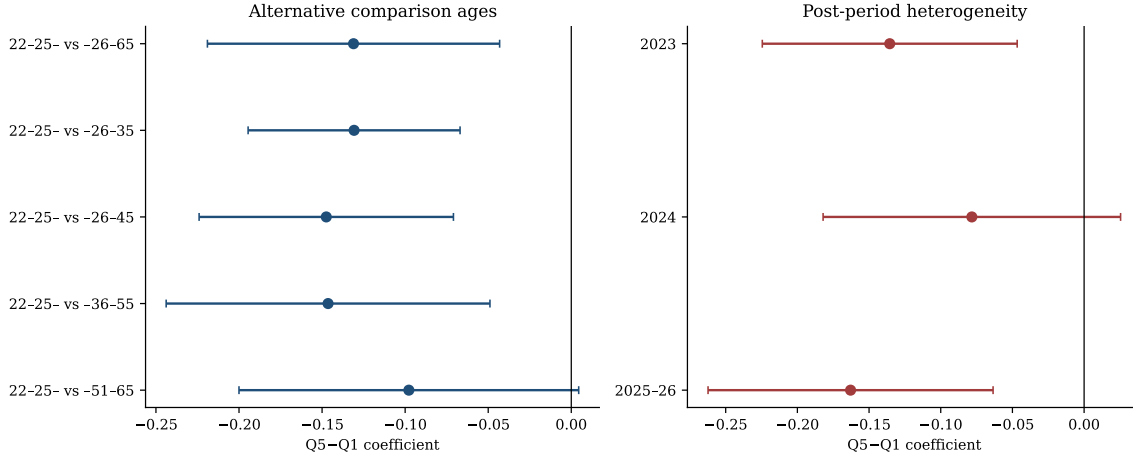
Occupation exclusions reinforce that reading without identifying mechanisms. Excluding food-preparation and serving occupations gives  $-0.1237$ ; computer/math gives  $-0.1158$ ; office/administrative support gives  $-0.1163$ . Excluding an approximate IND1990 leisure-and-hospitality set yields  $-0.1325$  with frozen membership and  $-0.1388$  after recomputing quintiles; code 800 also contains theaters and motion pictures. Deleting fast-food workers alone attenuates the original estimate to  $-0.1106$ . These are sensitivities, not estimates of a causal contribution.

Across the six selected measures, only 46 occupations are always Q1 and 18 always Q5.

They represent 7.34 and 2.41 percent of employment. The 271 occupations reclassified between tails and middle represent 59.19 percent of employment and 82.30 percent of mean fitted influence. Common support thus fixes composition but not the treatment groups supplying the estimate.

### 5.3 Age and time

Figure 4 makes comparison-age and era dependence explicit. The 22–25 versus 26–35 estimate,  $-0.1307$ , is virtually identical to the pooled 26–65 estimate; their paired difference is  $+0.0003$ . Estimates versus ages 26–45 and 36–55 are  $-0.1475$  and  $-0.1464$ . Against ages 51–65 the estimate is smaller,  $-0.0978$ , and differs from the pooled contrast by  $+0.0333$  with interval  $[0.0024, 0.0642]$ . In the separate six-bin profile, ages 31–40 have a positive  $+0.0645$  coefficient relative to 51–65. Prime-age evolution matters relative to the oldest comparison group, but the nearest 26–35 group does not explain away the pooled result.



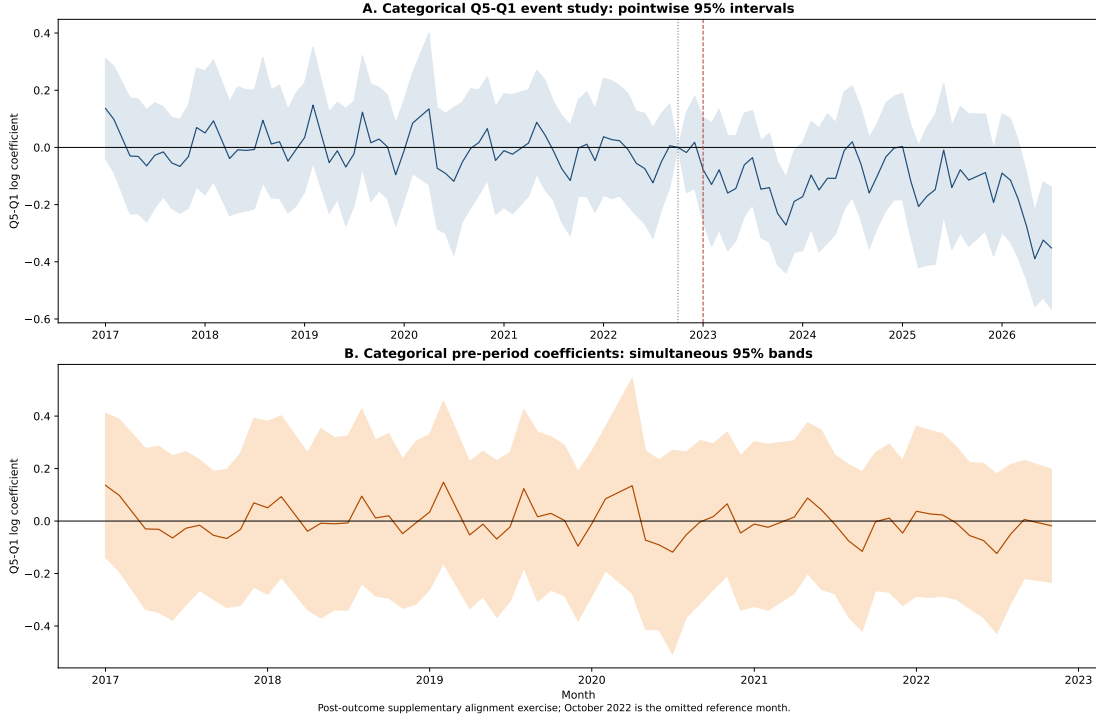
**FIGURE 4.** Comparison-age and post-period heterogeneity

*Notes:* Points are beta Q5–Q1 coefficients; lines are 95-percent common-multiplier wild-score intervals. Age models change the comparison stock and are not additive decompositions. Era estimates come from one joint model. All are post-outcome exploratory.

The 18–21 versus 51–65 estimate is  $-0.1050$  with interval  $[-0.2251, 0.0151]$ . Ages 22–25 are retained for comparability and because school-to-work selection is less severe than at 18–21, not because entry occurs at one universal age.

Time heterogeneity is nonmonotone: estimates are  $-0.1356$  in 2023,  $-0.0783$  in 2024, and  $-0.1629$  in 2025–26. The last differs from 2024 by  $-0.0846$  (interval  $[-0.1577, -0.0115]$ ), but not from 2023. The post-2025 extension therefore renews the contrast relative to 2024 rather than creating an unprecedented break. The original event study remains central: it detects no

differential pretrend under its common-multiplier max statistic, but absence of a detected pretrend does not identify AI against contemporaneous occupational shocks.



**FIGURE 5.** Categorical beta Q5–Q1 event study

*Notes:* Month-specific young-relative Q5–Q1 coefficients under the frozen design. December 2022 is the omitted transition month; October 2025 is absent. The figure is descriptive and does not turn January 2023 into an instrument.

Finally, adding Dingel–Neiman telework feasibility in a categorical joint-support model leaves beta at  $-0.1077$  (interval  $[-0.1934, -0.0220]$ ) while the remote coefficient is  $+0.0065$  (interval  $[-0.0214, 0.0343]$ ). For alpha, the exposure coefficient is  $-0.0808$  (interval  $[-0.1614, -0.0003]$ ) and the remote coefficient is  $-0.0132$  (interval  $[-0.0371, 0.0108]$ ). The remote sign changes with architecture; neither coefficient identifies a causal channel.

## 6 Continuous and Family Reparameterizations

The categorical results consume changing tail memberships. A continuous exercise asks a different question: conditional on Webb software exposure, how do the two selected families divide residual occupational variation? Let  $A$  be the equal-weight centroid of three standardized AIOE implementations and  $E$  the analogous Eloundou centroid. Define raw coordinates  $F = (A + E)/2$  and  $G = (A - E)/2$ , then standardize  $F$  and  $G$  separately in the implemented model.  $F$  is family-balanced consensus and  $G$  family disagreement; neither is a latent true

exposure.

The original exploratory model estimates  $b_F = -0.0404$  with interval  $[-0.0674, -0.0133]$  and  $b_G = +0.0309$  with interval  $[0.0029, 0.0589]$ . The reported original exploratory  $G$  tail area was .040; the identical refit under a new seed gives .032, reconciling rather than replacing the earlier number.

The requested leave-one-measure outcome refits change the interpretation. Removing alpha leaves  $F = -0.0330$  and  $G = +0.0342$ , with both intervals excluding zero. Thus the original asymmetry is not alpha-only. Removing any AIOE implementation also leaves the  $G$  interval above zero. But removing beta yields  $G = +0.0287$  with interval  $[-0.0009, 0.0584]$ , and removing broad yields  $+0.0215$  with interval  $[-0.0080, 0.0511]$ . The disagreement result is therefore materially sensitive to which task-family implementations receive equal family weight.

A representative one-AIOE model is more transparent: entered jointly, administrative AIOE is  $+0.0449$  (interval  $[0.0012, 0.0885]$ ) and beta is  $-0.0788$  (interval  $[-0.1292, -0.0284]$ ). A direct primitive model estimates  $D = -0.0323$  and  $S = -0.0303$ , with both intervals below zero. These are alternative parameterizations and conditioning exercises, not independent corroborations.

Because the fitted regressors are  $F/s_F$  and  $G/s_G$ , the exact original-unit family transformation is

$$\gamma_A = \frac{1}{2} (b_F/s_F + b_G/s_G), \quad \gamma_E = \frac{1}{2} (b_F/s_F - b_G/s_G). \quad (4)$$

Transforming every common draw consistently gives  $\gamma_A = +0.0248$  (paired SE 0.0199) and  $\gamma_E = -0.0707$  (paired SE 0.0272). This is an exact change of basis of the same fitted predictor, not a second outcome model. The raw  $F$  and  $G$  weighted correlation is 0.218 in the original construction and varies under deletion; standardization does not imply orthogonality.

The  $D + \lambda S$  grid reaches the same caution from within the task family. All five categorical Q5–Q1 estimates are negative, from  $-0.1013$  at  $\lambda = 0$  to  $-0.1482$  at  $\lambda = 1$ , while fixed-normalization and restandardized continuous coefficients differ. The family evidence is best summarized as a negative consensus association plus construction-sensitive conditional asymmetry. It does not rank exposure families universally.

## 7 Occupational Mobility and Ranking Statements

### 7.1 Linked switches and two threshold concepts

The CPS rotation design supplies exact adjacent-month links for valid months-in-sample; the eight-month gap from month four to five is excluded. December 2019–January 2020 is excluded symmetrically where a destination code crosses taxonomies. Origins use longitudinal weights. The all-six frame contains 108,500 employed-to-employed occupational switches; it does not contain entrants from nonemployment.

At threshold zero, an architecture assigns a switch a positive, negative, or tied direction. Six-way directional conflict—at least one positive and one negative label—is 53.28 percent by official weight. Movement-mass weighting, with mass equal to the sum of absolute standardized score movements across architectures, lowers conflict to 40.10 percent. The no-alpha set has 35.02 percent zero-threshold conflict and 15.35 percent mass-weighted conflict; within AIOE the corresponding shares are 11.35 and 2.10 percent. Changing the number and composition of possible pairs is part of those comparisons.

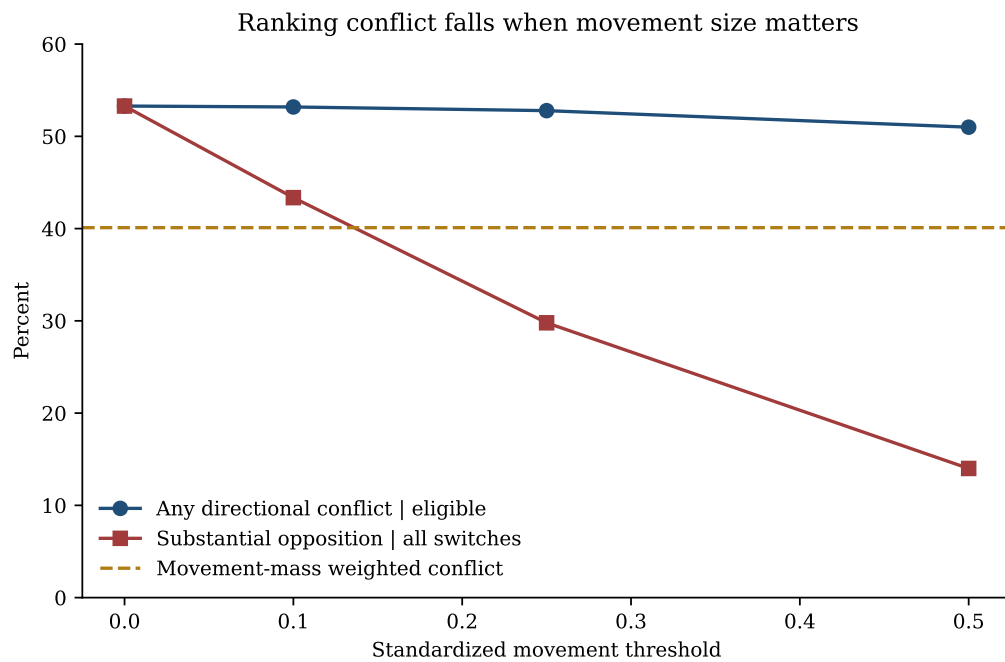
Figure 6 distinguishes two referee-requested threshold definitions. A material-move restriction first requires at least one architecture to exceed the threshold and then computes any directional conflict. Substantial opposition requires both a positive and a negative movement to exceed it. On all switches, substantial opposition falls from 53.28 percent at zero to 43.35, 29.79, and 14.00 percent at 0.1, 0.25, and 0.5 standard deviations. At 0.25 SD, 97.42 percent of switch weight is eligible and conditional substantial opposition is 30.58 percent. Percentile-rank thresholds deliver the same qualitative decline.

Immediate A–B–A reversals have 55.15 percent zero-threshold conflict; persistent A–B–B changes have 54.46 percent. At 0.5 SD, substantial-opposition shares are 13.57 and 14.39 percent. Persistence therefore does not make conflict disappear, and an immediate reversal is not classified as a coding error. Named nontrivial disagreements include accountants to bookkeeping clerks, for which AIOE falls while alpha and beta rise; registered nurses to medical and health-services managers, for which AIOE rises and alpha falls; and retail salespersons to cashiers, where AIOE falls, alpha rises, and beta/broad fall.

### 7.2 Aggregate direction and the hard benchmark

The disagreement has limited aggregate consequences in this frame. Across the six architectures, 49.21–50.49 percent of switch weight moves toward higher measured exposure, and 20.0–23.4 percent lands in the destination top weighted-rank quintile. Architecture choice changes





**FIGURE 6.** Directional conflict depends on movement scale

*Notes:* Blue conditions any conflict on the R2-style material-move eligible set. Red reports R1-style substantial opposition as a share of all switches. The dashed line is conflict weighted by total absolute standardized movement. All use official longitudinal weights.

individual labels more than the aggregate upward/downward balance. Entrant destination classifications cannot be computed from the switch-only frame.

The hard rematching benchmark preserves age-by-month, broad origin and destination families, and detailed origin/destination marginals while destroying the detailed pairing. On 30,170 strata, 200,000 pseudo-units, and 999 draws, it represents 98.31 percent of switch weight and gives expected conflict 52.32 percent (across-rematch SD .00054; Monte Carlo SE of the mean .000017). The realized 53.28 percent is 0.96 percentage points higher. This is a near-threshold sample difference, not equivalence or proof of negligibility.

The benchmark now prohibits identical detailed origin and destination codes and repairs assignments while preserving stratum margins; no false self-switches remain. The exact product-marginal expectation, 50.29 percent, permits self-transitions with probability 3.66 percent and therefore does not apply to the constrained null. Pseudo-size checks at 50,000 and 100,000 units represent 73.86 and 95.30 percent of weight and leave gaps of 0.92 and 0.91 points. The conclusion is that broad assortativity explains most excess conflict, while the exact magnitude remains benchmark-dependent.

Earlier flow regressions for employment exit, occupational outflow, and entry destination were imprecise and use different risk sets. They do not add to the stock coefficient and are retained in the appendix rather than promoted into a stock-flow decomposition.

## 8 Synthesis, Limits, and Conclusion

The application yields three levels of conclusion. First, construction choices demonstrably change the empirical treatment. Taxonomy routes determine which occupations enter; external concepts change tail memberships and coefficients; the Q1 reference supplies a material share of the original contrast; and comparison-age composition changes the magnitude relative to the oldest workers. These are not generic caveats. They are measured links between upstream choices and downstream claims.

Second, some narrower statements survive declared perturbations. The primary beta coefficient remains negative after calendar repair, reasonable minimum-size thresholds, stable-taxonomy and post-2020 comparisons, food-service, computer/math, office-support, and approximate leisure/hospitality exclusions. Within the selected six implementations, every common-support point estimate is negative. The task-family  $D + \lambda S$  grid is also uniformly negative. Those statements must carry their architecture set, support, reference tail, age comparison, and descriptive interpretation.

Third, the stronger claims do not survive. The predeclared all-six simultaneous band fails.

Outside architectures are smaller and imprecise. Q5 is not detectably different from Q2 or Q4. A within-family permutation does not demonstrate AI specificity. F/G asymmetry becomes imprecise when beta or broad is removed. Mobility conflict is frequent at zero but declines with economically meaningful thresholds and does not change aggregate moving-higher shares much. The evidence therefore does not justify a causal AI effect, a universal family ordering, a monotone dose response, or general occupational-rank interchangeability.

The contrast with prior labor-market evidence is consequently bounded. ADP payroll data can measure firm and hiring margins unavailable in CPS (Brynjolfsson, Chandar, and Chen 2026). Danish adoption-linked administrative data can estimate earnings and hours effects conditional on reported use (Humlum and Vestergaard 2025). This paper shows that changing exposure construction inside one CPS stock design changes the apparent conclusion. It cannot determine which difference in data, country, outcome, adoption, or architecture explains disagreements across those studies.

Conventional uncertainty is incomplete as well. The reported intervals condition on selected exposure labels, mappings, and support; no validation sample reveals a true latent exposure. One- and two-way clustered checks do not reconstruct the full CPS survey design. Realized precision was more than three times worse than prospective calculations, and the available simulations do not identify why. These limits reduce confidence rather than supplying a preferred interpretation.

The resulting methodological lesson is simple. A constructed exposure should be reported as an architecture: technology scope, primitive, label source, aggregation, taxonomy, support, normalization, and representation. Common support is useful but does not make ranks or tail membership common. An aggregate coefficient can be stable within a selected architecture family while an external architecture, reference group, or mobility threshold changes the economic statement.

The durable conclusion is thus measurement-centered. The early-career CPS pattern is a useful stress test, not an identified AI displacement estimate. Robustness belongs to a specified statement and construction, not to the name of the variable.

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